

# Felipe Santos-Marquez

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## EDUCATION

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### Technische Universität Dresden

Doctor rerum politicarum (Dr. rer. pol.) in Economics

– Dissertation: “Essays on Regional and Development Economics”

– Grade: *Summa Cum Laude*

Dresden, Saxony, Germany

April 2021–March 2026

### University of Oxford, Lady Margaret Hall

Summer program: Advanced Applications of Neural Networks and Deep Learning

Oxford, UK

July–August, 2023

### Nagoya University

MA in International Development

Nagoya, Aichi, Japan

2019–2021

– Thesis: “Achieving Peace and the SDGs in Colombia: Municipal Convergence, Crime, and the Role of Conditional Cash Transfers”

### Tafe institutions EAA & AITM

Certificate III and IV in hospitality and Management (commercial cookery)

Melbourne & Brisbane, Australia

2013–2014

### Universidad Nacional de Colombia

5-year Pregrado in Physics - BSc (Physics)

Bogota, Colombia

2007–2012

## ACADEMIC AND RESEARCH EXPERIENCE

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### Technische Universität Dresden

Research Associate at the Chair of Economics esp. International Economics

Dresden, Saxony, Germany

04/2021- 12/2025

### Nagoya University

Research Assistant at the QuaRCS lab

Nagoya, Aichi, Japan

04/2020- 08/2020

### Nagoya University

Academic Tutor at the Graduate School of International Development

Nagoya, Aichi, Japan

09/2019- 02/2021

### National University of Colombia

Mathematics (Topology and Analysis) tutor at the School of Mathematics, Faculty of Science

Bogota, Colombia

01/2008- 07/2008

## TEACHING EXPERIENCE

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- **Guest lecturer** at Nagoya University January 2025  
*AI and Deep Learning*
- **Technische Universität Dresden** winter 2021 - present  
*Empirical Economics course programming and replication tutorials - introduction to R - AI and Deep Learning in Python - international trade: theory and policy Tutorials*
- **Guest lecturer** at Nagoya University summer 2022

- **Teaching Assistant** at Nagoya University fall 2020  
*GSID Open Webinar Series and Spanish*
- **English Teacher** at Alive English School, Nagoya city, Japan 2017-2021  
*Preschool, elementary and secondary classes*

## PUBLICATIONS (PEER-REVIEWED ARTICLES)

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- [1] A. Domínguez, **F. Santos-Marquez**, and D. Castells-Quintana, “The geography of energy transitions: A network approach for post-fukushima japan”, *Energy Economics*, vol. 151, p. 108911, 2025, ISSN: 0140-9883.
- [2] C. Mendez and **F. Santos-Marquez**, “Economic and social disparities across subnational regions of south america: A spatial convergence approach”, *Comparative Economic Studies*, pp. 1–24, 2022, Available at <https://doi.org/10.1057/s41294-021-00181-0>.
- [3] A. Domínguez, **F. Santos-Marquez**, and C. Mendez, “Sectoral productivity convergence, input-output structure and network communities in japan”, *Structural Change and Economic Dynamics*, 2021, Available at <https://doi.org/10.1016/j.strueco.2021.10.012>.
- [4] **F. Santos-Marquez**, “Spatial beta-convergence forecasting models: Evidence from municipal homicide rates in colombia”, *Journal of Forecasting*, 2021, Available at <https://doi.org/10.1002/for.2816>.
- [5] **F. Santos-Marquez**, A. B. Gunawan, and C. Mendez, “Regional income disparities, distributional convergence, and spatial effects: Evidence from indonesian regions 2010–2017”, *GeoJournal*, pp. 1–19, 2021, Available at <https://doi.org/10.1007/s10708-021-10377-7>.
- [6] C. Mendez and **F. Santos-Marquez**, “Regional convergence and spatial dependence across subnational regions of asean: Evidence from satellite nighttime light data”, *Regional Science Policy & Practice*, 2020, Available at <https://doi.org/10.1111/rsp3.12335>.
- [7] **F. Santos-Marquez** and C. Mendez, “Regional convergence, spatial scale, and spatial dependence: Evidence from homicides and personal injuries in colombia 2010-2018”, *Regional Science Policy & Practice*, 2020, Available at <https://doi.org/10.1111/rsp3.12356>.

## MASTER THESIS SUPERVISION (TU DRESDEN)

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- Lisa-Marie Muller (Dresden, 2021)  
Tracing Human Light Investigation of low-intensity nightlight satellite data in rural areas of Rwanda
- Arik Dinesen (Dresden, 2022)  
The Economic Impact of Large scale Oil Discovery in Norway
- Kyra F. Riederer (Dresden, 2022)  
The effects of the New Rural Pension System on labour supply in China
- Paavo Maria Brummer (Dresden, 2023)  
Mobile Internet Coverage and Its Effects on Education Outcomes in Colombia
- Paul Fischer (Dresden, 2023)  
Network-centrality and market access of cities – an empirical analysis
- Isabel Knippel (Dresden, 2024)  
The effect of the automatic information exchange on economic activity of offshore financial centers – a view from outer space

## CONFERENCE & WORKSHOP PRESENTATIONS

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- EEA-ESEM  
*Rotterdam, Netherlands* September –2024
- Japanese Economic Association, spring meeting  
*Tokyo, Japan* May –2024
- 20th World Congress International Economic Association, EAFIT  
*Medellin, Colombia* December –2023
- 62nd ERSA Congress, University of Alicante  
*Alicante, Spain* August–2023
- 4th Swiss Workshop on Local Public Finance and Regional Economics, University of Bern  
*Bern, Switzerland* October–2022
- 1st International Workshop on the Chinese Development Model, IQS School of Management  
*Barcelona, Spain* July–2022
- Young Scholars Initiative meeting at International University College of Turin  
*Torino, Italy* June–2022
- 60th Annual ERSA Congress “Territorial Futures – Visions and Scenarios for a resilient Europe”  
*Bolzano, Italy* August–2021
- ERSA Summer School 2021 - Spatial Analysis of Regional Inequalities  
*Groningen, Netherlands* June-July–2021
- XV World Conference of Spatial Econometrics Association  
*Tokyo, Japan* May–2021
- 57th Conference of the Japan Society of Social Science on Latin America  
*Nagoya, Japan* November–2020
- Bolivian Conference on Development Economics  
*La Paz, Bolivia* August–2020
- Applied Regional Science Conference  
*Saga University, Saga, Japan* November–2019

## RESEARCH VISITS

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- Graduate School of International Development, University of Nagoya, Japan About twice a year from –2021
- Asian Growth Research Institute (AGI), Kitakyushu, Japan September –2022
- AQR Research Group, University of Barcelona, Spain July –2021
- Universidad de los Andes, Bogota, Colombia December –2019

## AWARDS

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- Prize of the Deutsche Bundesbank for support for international research cooperation June –2021
- Travel award for a workshop given by the TU Dresden Graduate Academy January –2023
- Travel award for a summer school given by the TU Dresden Graduate Academy July –2023

- Travel award for transportation expenses for a summer school given by the Association of Friends and Sponsors of TU Dresden e. V. July –2023
- Lecture award given by the German Economic Association (Verein für Socialpolitik) December –2024

## COMPUTER SKILLS

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- **Programming:** R, Python, c++, Stata and Matlab
- **User:** GitHub, Geoda and QGIS

## LANGUAGES

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- **Spanish:** Native proficiency level
- **English:** Native proficiency level
- **Japanese:** Limited working proficiency
- **German:** Limited proficiency

## FIRM VISITS

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As part of the Deep Learning course, I organized student visits to firms in the semiconductor industry and Silicon Saxony to show students how AI methods are used in industry.

- Student visit to Bosch Dresden January –2025  
[https://tu-dresden.de/bu/wirtschaft/vwl/iwb/die-professur/news/student-visit-to-bosch-dresden?set\\_language=en](https://tu-dresden.de/bu/wirtschaft/vwl/iwb/die-professur/news/student-visit-to-bosch-dresden?set_language=en)
- Student visit to Infineon Dresden February –2024  
[https://tu-dresden.de/bu/wirtschaft/studium/news-1/student-visit-to-infineon-dresden?set\\_language=en](https://tu-dresden.de/bu/wirtschaft/studium/news-1/student-visit-to-infineon-dresden?set_language=en)

## EXTRACURRICULAR AND ACADEMIC ACTIVITIES

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- Data@ANZ ANZ virtual experience Program August–2020  
*Completed two practical task modules in*  
Exploratory Data Analysis  
Predictive Analytics.
- DataCamp courses- completed several online courses in: 2019 –present  
*Cluster Analysis in R*  
Supervised Learning in R: Classification  
Unsupervised Learning in R  
Spatial Analysis with sf and raster in R.  
Visualizing Geospatial Data in R  
Intermediate Python  
Network Analysis in R  
Introduction to Data Visualization with Matplotlib  
Machine Learning in the Tidyverse  
Introduction to Functions in Python  
Introduction to Data Visualization with Seaborn  
Fundamentals of Bayesian Data Analysis in R  
Supervised Learning with scikit-learn  
Introduction to Deep Learning with Pytorch